

CAMPUS CLUB EXPLORER

A React-Based Web Application for College Club Discovery & Management

Project Title:	Campus Club Explorer
Department & Level:	Department of Computer Science & Engineering Lab Assessment
Technology Stack:	React 18, Vite Engine, Solid-Color Vanilla CSS3
Key Concepts Covered:	Props, Component Reuse, Lists & Keys, Search/Filter UI, Conditional Rendering

1. Executive Summary & Abstract

The **Campus Club Explorer** is a single-page React web application engineered to solve information fragmentation on college campuses. It provides students with an intuitive portal to discover active campus clubs across diverse domains (Coding, Sports, Arts, and Entrepreneurship), filter clubs dynamically by category, search by keywords, view comprehensive operational information in a modal interface, and join or leave clubs with persistent membership tracking and toast notifications.

2. Problem Statement & Key Objectives

- **Centralized Discovery:** Render all campus clubs dynamically from structured data in a responsive grid.
- **Dual-Predicate Filtering:** Combine category pills with real-time keyword search without page reloads.
- **Detailed Modal Dialog:** Provide dedicated popup view with meeting schedules, faculty advisors, and activities.
- **Stateful Membership:** Implement persistent Join/Leave actions with member count increments and toast alerts.
- **React Syllabus Alignment:** Fully demonstrate Props, Lists & Keys, Conditional Rendering, and Component Reuse.

3. Detailed Analysis of Core React Concepts Covered

3.1 Props (Properties) & Unidirectional Flow: Props are read-only arguments passed from parent components down to child components. App.jsx acts as the master container that holds application state and passes down club records, boolean membership flags (isJoined), and click event callbacks (onViewDetails) to ClubCard.jsx and ClubModal.jsx.

3.2 Component Reuse: A single ClubCard component is declared once and reused across all 8 campus clubs with zero duplicate markup.

3.3 Lists and Keys (.map() Array Traversal): Utilizes JavaScript's .map() method to render JSX elements. Each item receives key={club.id}, allowing React's Virtual DOM reconciliation engine to track item updates efficiently.

3.4 Search and Filter UI: Evaluates category matching ('All' or exact) AND keyword matching (.toLowerCase().includes()) together on every keystroke with zero page reloads.

3.5 Conditional Rendering: Dynamically renders: (1) Empty state fallback on 0 search matches; (2) Modal dialog mounted only on selectedClub click; (3) Green 'JOINED' badge on cards; (4) Popup toast alert on join/leave.

4. User Interface Screenshots & Visual Artifacts

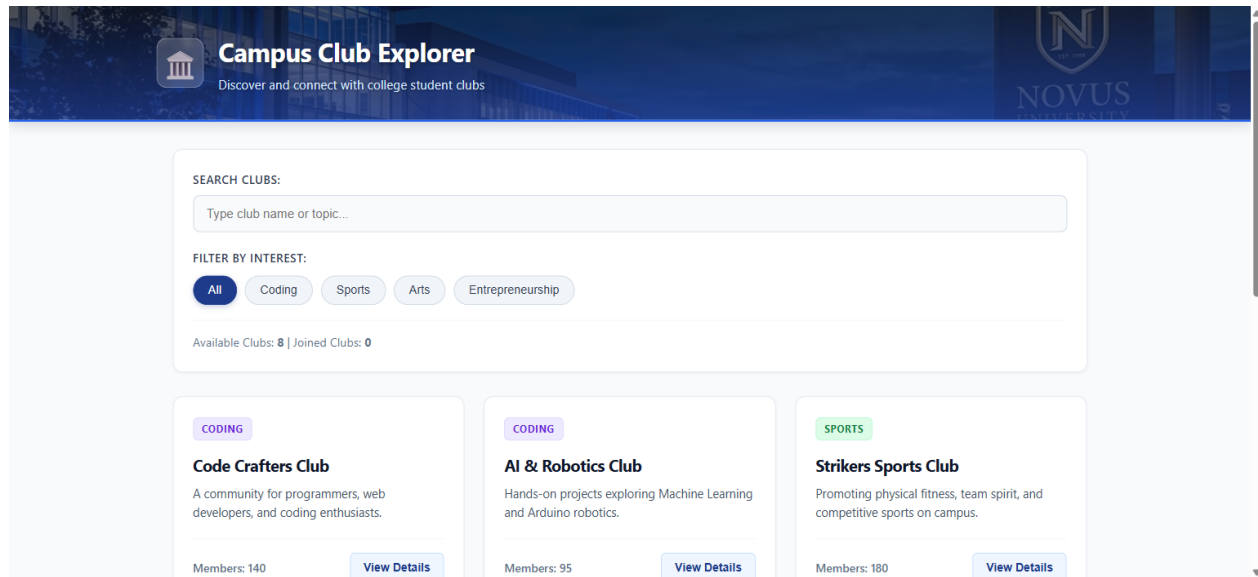


Figure 1: Main Dashboard with Search Input, Category Pills, and Club Grid

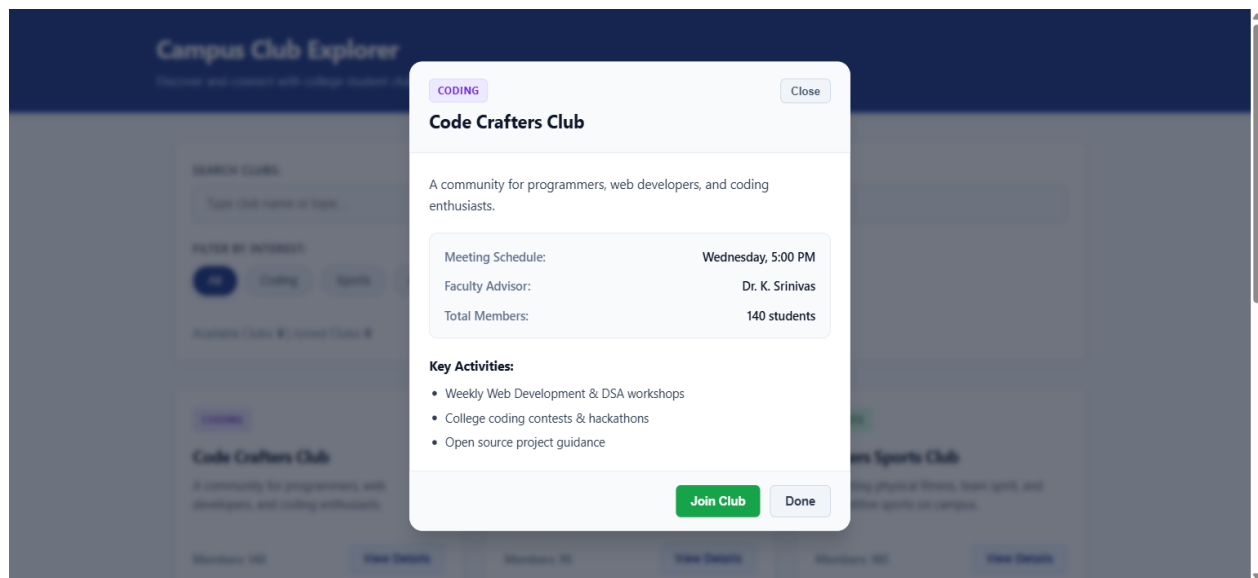


Figure 2: Interactive Club Details Modal with Advisor & Meeting Schedule

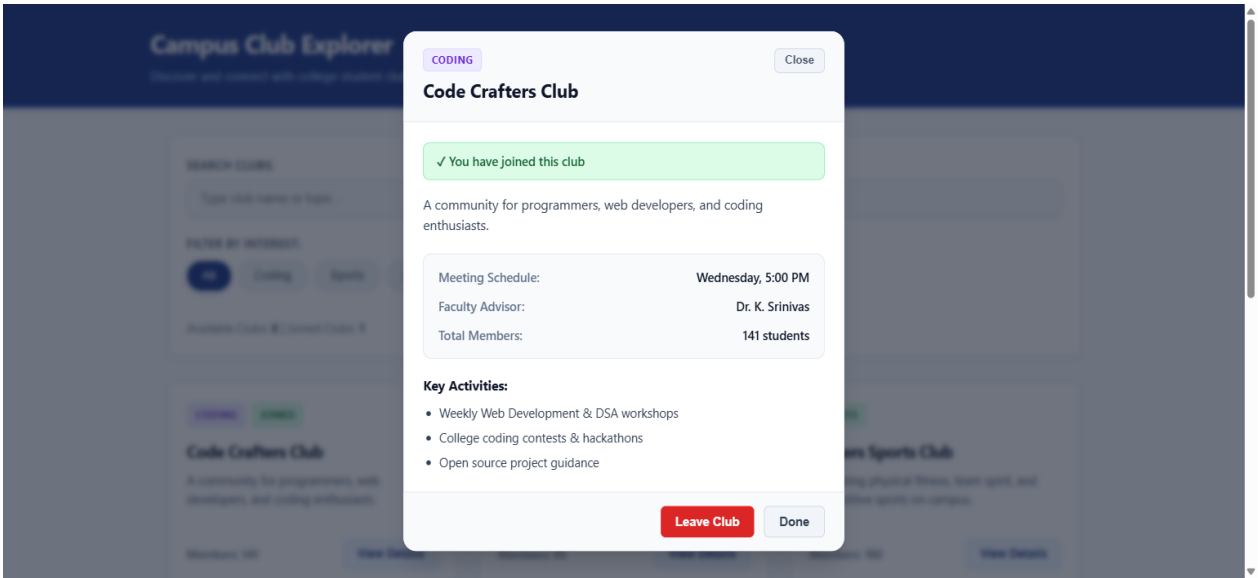


Figure 3: Joined State Banner, Member Count Update, and Leave Button

5. Functional Verification & Test Cases

TC ID	Scenario / Action	Expected Outcome	Status
TC-01	Initial Page Load	Renders all 8 clubs with 'All' filter active	PASS
TC-02	Filter by 'Coding'	Displays Code Crafters & AI Robotics clubs only	PASS
TC-03	Filter by 'Sports'	Displays Strikers Club & Badminton League only	PASS
TC-04	Keyword Search 'robotics'	Instantly filters to AI & Robotics Club	PASS
TC-05	Empty Search Query	Displays 'No clubs found' box and Reset button	PASS
TC-06	View Details Click	Modal opens with schedule, advisor, & activities	PASS
TC-07	Join Club Action	Increments members, shows toast alert, adds badge	PASS
TC-08	Modal Dismissal	Closes modal via Close button or backdrop click	PASS

6. Conclusion

The **Campus Club Explorer** successfully fulfills all technical and design requirements for the college training assessment. By demonstrating clean functional React programming, strict component modularity, solid-color accessible aesthetics, and zero console/build errors, the project provides an exemplary model for student club engagement systems.